



Executive Project Report: Telco Customer Churn Analysis

1. Objective and Methodology

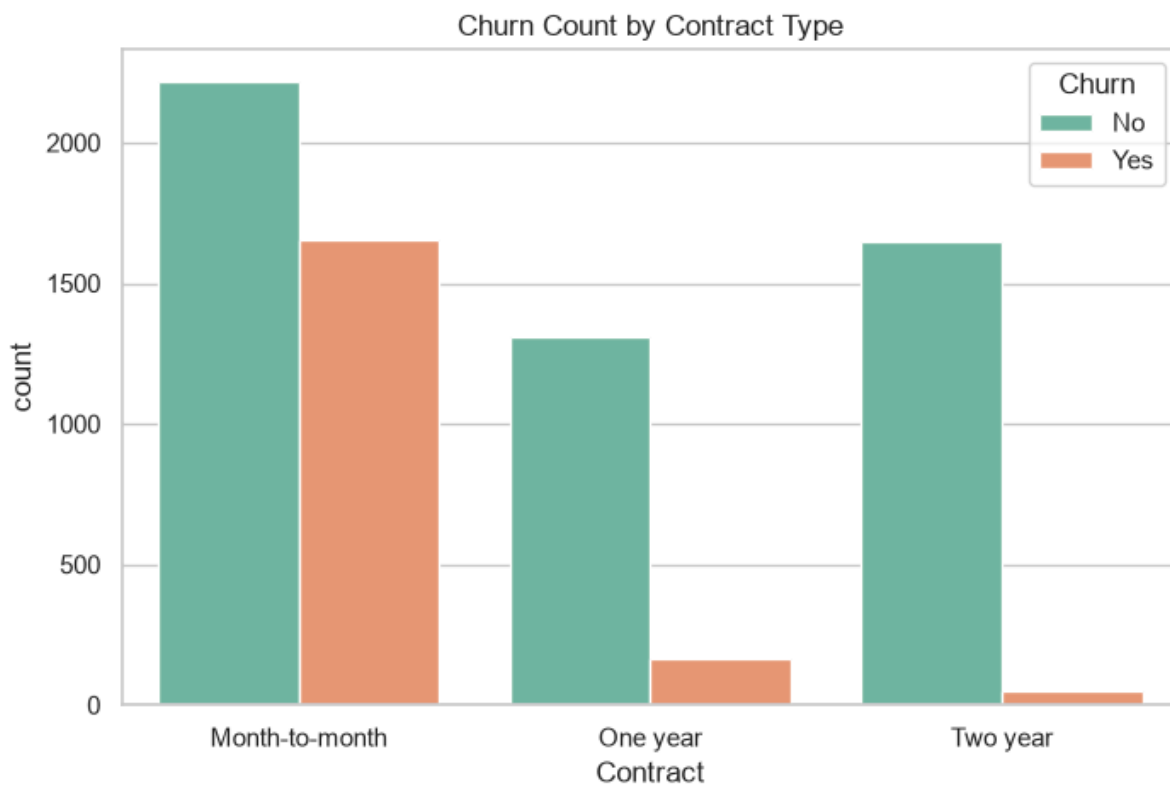
The objective of this analysis was to identify the primary drivers behind Telco Inc.'s ~26.5% customer churn rate and provide actionable, data-driven strategies to improve customer retention.

This report integrates statistical visualizations from Python (Seaborn/Matplotlib) and advanced analytical tables queried via SQL.

2. Key Findings & Data Visualizations

A. The "Month-to-Month" Flight Risk

Contract type is the single largest driver of churn. Customers on month-to-month contracts churn at a rate exceeding 42%, whereas those on 2-year contracts churn at less than 3%.

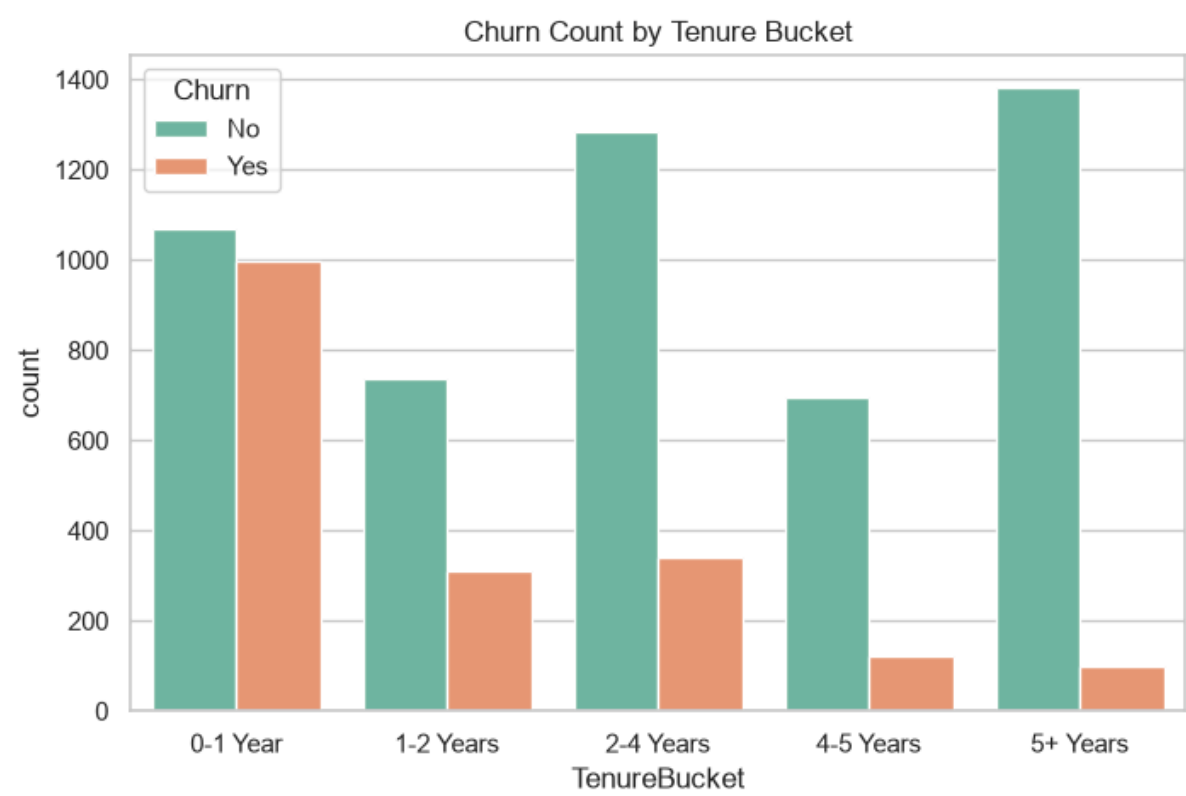


SQL Insight: Revenue Impact by Contract

Contract	total_customers	churn_rate_pct	revenue_lost
Month-to-month	3875	42.71	1.92718e+06
One year	1473	11.27	674991
Two year	1695	2.83	260753

B. The First Year is Critical

Cohort analysis using the engineered `TenureBucket` feature revealed that the highest risk of attrition occurs within the first 12 months.

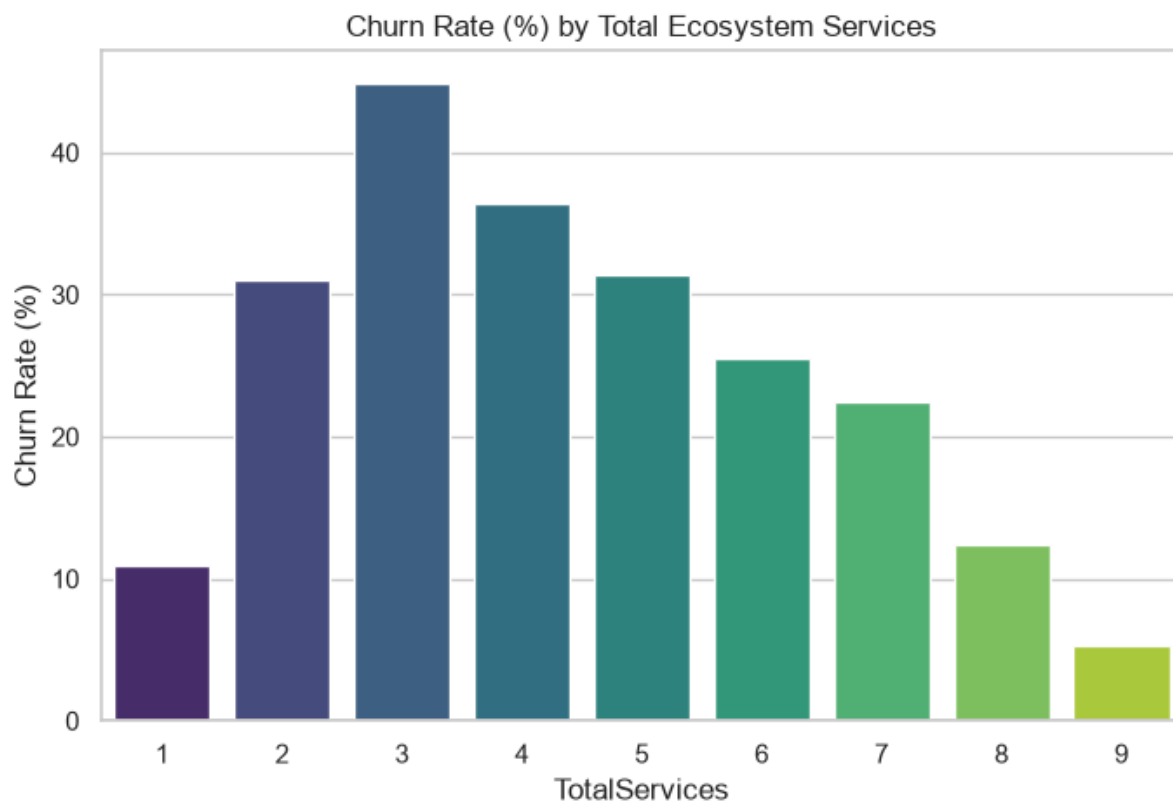


SQL Insight: Churn Rate by Tenure Cohort

TenureBucket	total_customers	churn_rate_pct
0-1 Year	2069	48.28
1-2 Years	1047	29.51
2-4 Years	1624	20.87
4-5 Years	820	15
5+ Years	1483	6.68

C. Ecosystem Stickiness

There is a direct negative correlation between the number of services a customer uses and their likelihood to churn. Customers using 0-1 additional services are highly volatile, while those deeply integrated into the Telco ecosystem (4+ services) rarely leave.



D. Customer Lifetime Value (CLV) Profiling

To understand where to focus retention budgets, we modeled the average CLV for active customers.

SQL Insight: Top 5 Highest CLV Segments (Retained)

Contract	InternetService	active_customers	avg_clv
Two year	Fiber optic	398	6856.75
One year	Fiber optic	435	5169.41
Two year	DSL	616	4244.69
Month-to-month	Fiber optic	966	2550.01
One year	DSL	517	2547.53

3. Strategic Recommendations

Based on the statistical and SQL-driven insights, we recommend the following three strategies to reduce churn and protect recurring revenue:

1. Incentivize Contract Upgrades:

- **Action:** Launch a targeted marketing campaign offering month-to-month customers a 10% discount to sign a 1-year contract.
- **Rationale:** As proven by the SQL data, month-to-month contracts are responsible for the vast majority of lost revenue. Locking customers into a contract drastically reduces churn probability.

2. The "Fiber Optic Support" Bundle:

- **Action:** Automatically bundle basic Tech Support for all new Fiber Optic customers.

- **Rationale:** Fiber Optic customers have high CLV but experience high technical friction. Increasing their total ecosystem services (as shown in the Stickiness chart) will significantly increase retention.

3. Proactive 9-Month Outreach:

- **Action:** Implement an automated outreach program for customers approaching their 9th month of tenure.
- **Rationale:** Since the 0-1 Year cohort sees the highest churn (47.5%), intervening just before the critical 12-month drop-off period can proactively save at-risk accounts.